

THE PIPELINE FACTORY

Ep 1: The factory

Day one.
Where do I start?

With the
big picture.



Ep 1

THE PROBLEM

I run every script by hand, one sample at a time.



Nextflow fixes that. It's a factory manager: every step, in order, for every sample. By itself.



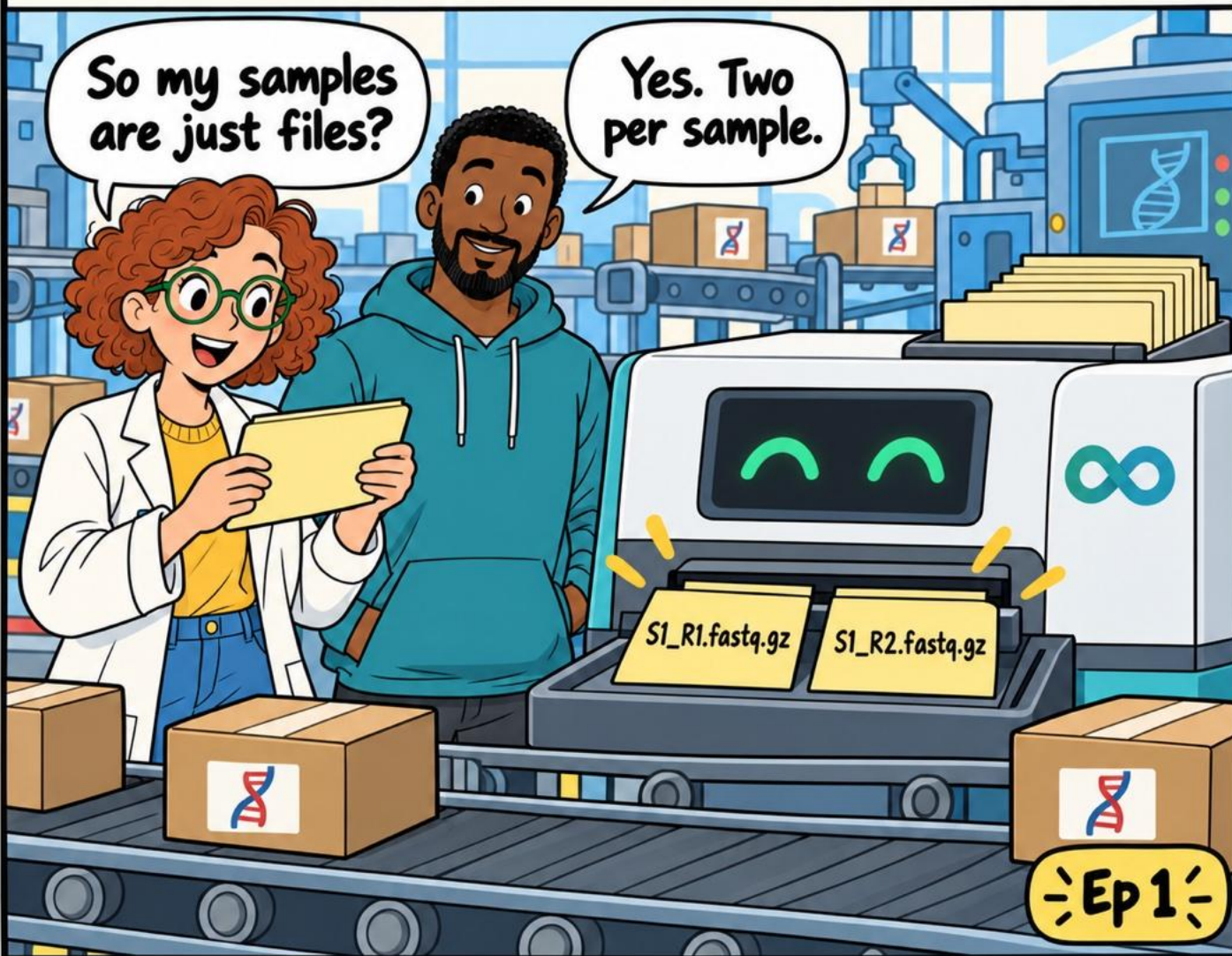
Ep 1

THE RAW MATERIALS

Reads = short pieces of RNA, read by a sequencing machine and saved in files.

So my samples are just files?

Yes. Two per sample.



THE FOLDER

What's all this?



main.nf:
blueprint,
stations in order

nextflow.config:
control room,
laptop or cluster?
memory?

bin/: toolbox,
your own scripts

data/: raw
materials,
your reads

Ep 1

WHY KEEP THEM APART?

Why not one big file?

New computer?
Change only the control room.

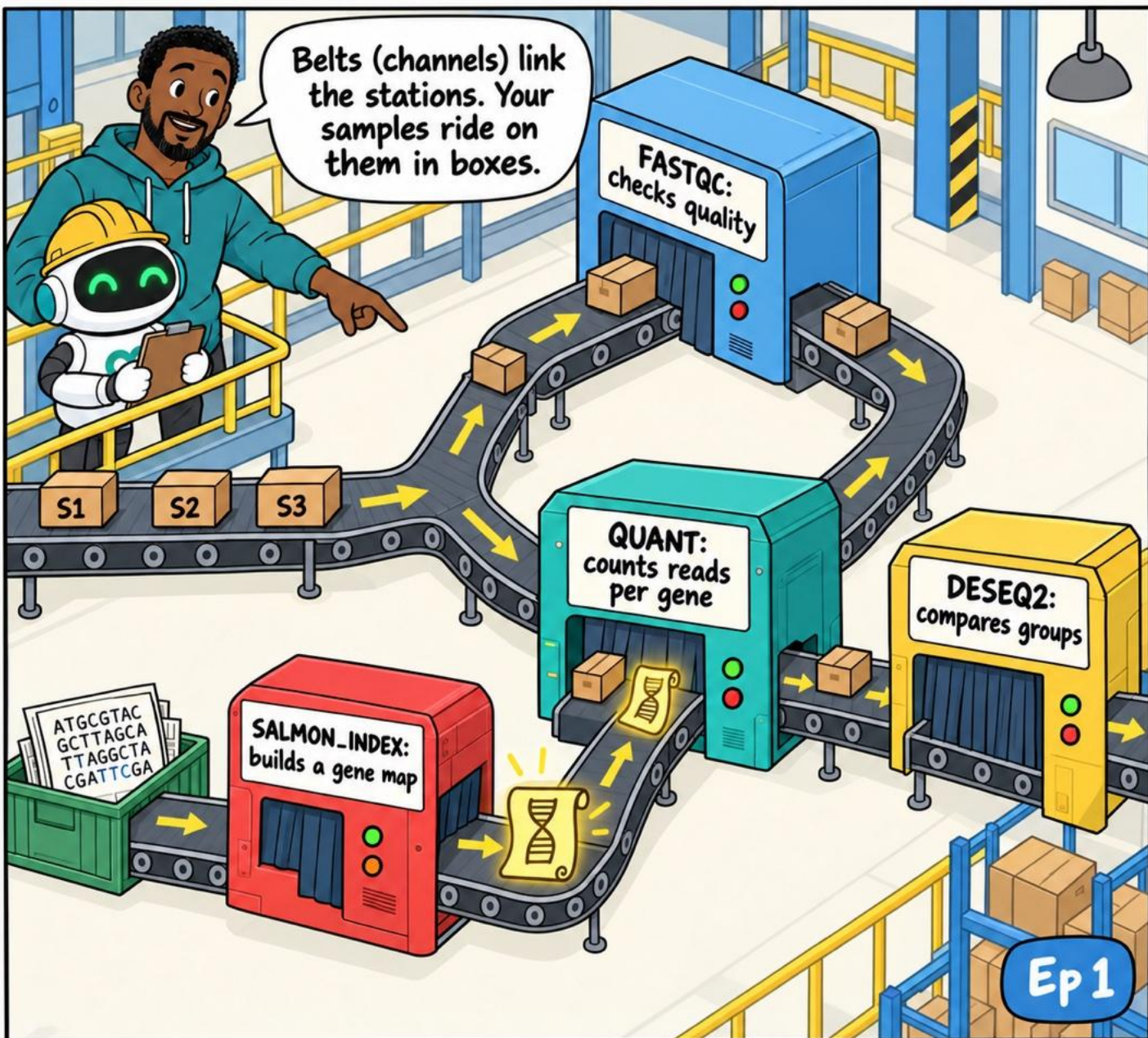
The blueprint stays the same.



Ep 1



THE STATIONS



THE FOLDER, IN CODE

my-rnaseq/

```
|— main.nf          # blueprint: stations + order
|— nextflow.config  # control room: where, how much power
|— bin/             # toolbox: your own scripts
|   |— run_salmon.sh
|   |— deseq2.R
|— data/            # raw materials
|   |— S1_R1.fastq.gz
|   |— S1_R2.fastq.gz
|   |— transcripts.fa.gz # gene sequences
|   |— samples.csv      # which group each sample is in
```

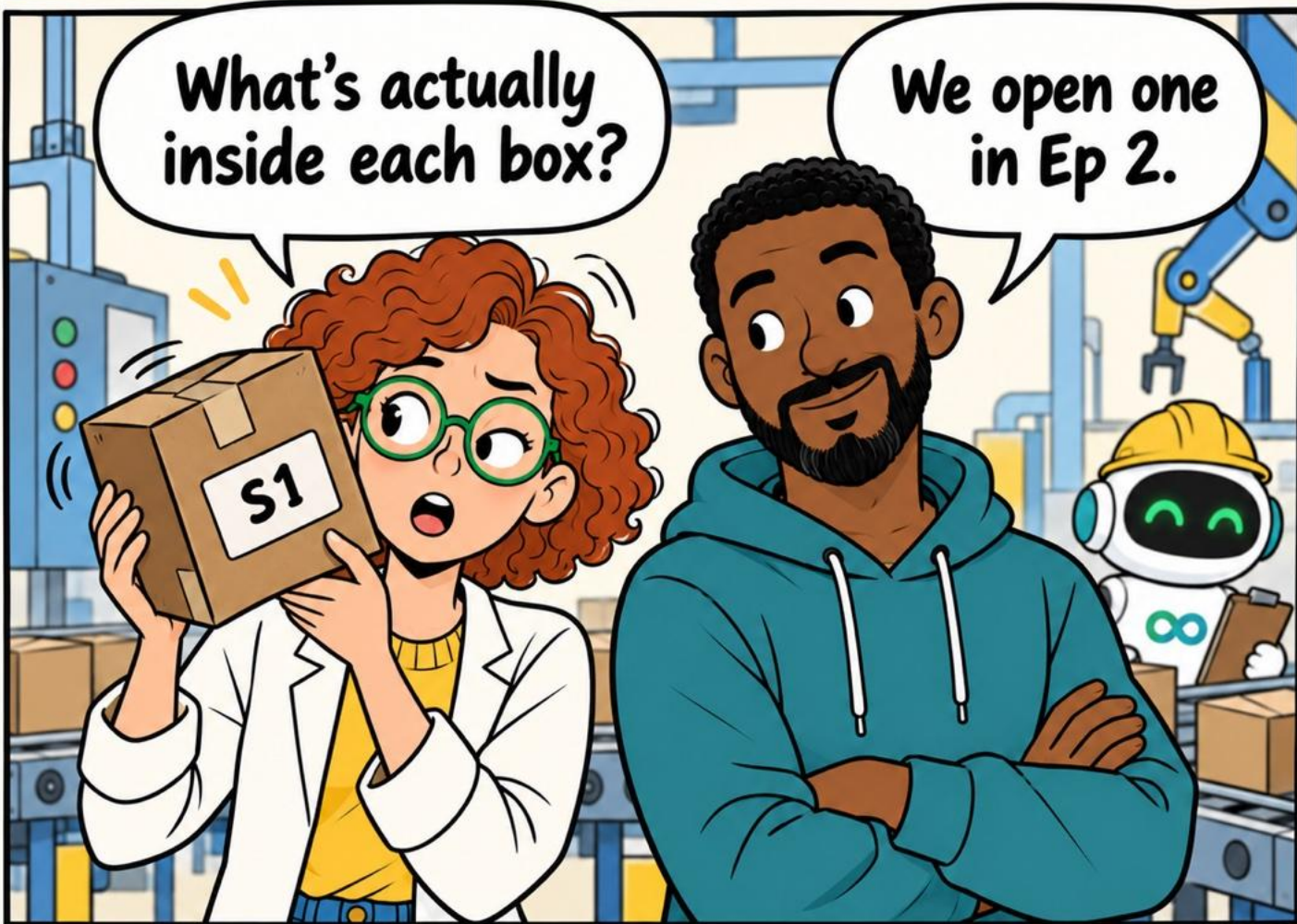


Ep 1

≡ NEXT TIME ≡

What's actually inside each box?

We open one in Ep 2.



≡ Ep 2: The box → ≡

Ep 1



THANKS FOR READING!

Follow me on GitHub,
where I explain how it all works:

<https://github.com/loukesio>